MAT367
Intro to Differential Geometry

Class Lecture Notes

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Preface

These notes were created during class lectures. As such, they may be incomplete or lacking in some detail at parts, and may contain confusing typos due to time-sensitivity. Additionally, these notes may not be comprehensive. Most statements in this document which are not Theorems, Problems, Lemmas, Corollaries, or similar, are likely paraphrased to a certain degree. Please do not treat any material in this document as the exact words of the original lecturer.

If you are viewing this document in Obsidian, you may notice that the links in the pdf document do not work. This is intentional behaviour, as I currently do not have or know of a decent solution which allows them to behave well with the setup in Obsidian. However, below certain pages, there may be links to other documents - these are usually context-relevant links between notes of different areas of study. I created these links to point out potential similarities, or in case one area of study is borrowing a concept, definition, or theorem from another area of study, and you wish to see the full, original definition/derivation/proof or whatever it may be.

IV Tangent Spaces and Vector Fields

7 Tangent Spaces

Lec 11 - Feb 10 (Week 6)

recall: if $M \subseteq \mathbb{R}^n$ an embedded mfld, the **tangent space** at $p \in M$ is T_pM given as:

$$T_pM \;=\; \{\text{velocity vectors of curves } \gamma : (a,b) \longrightarrow M \text{ at } 0 \text{ where } \gamma(0) = p\} \subseteq \mathbb{R}^n$$

this is in fact an m-dim'l vspace. indeed, M^m is locally the img of some $\varphi : U \subseteq \mathbb{R}^m \longrightarrow \mathbb{R}^n$, so $T_pM = D\varphi\big|_{\varphi^{-1}(p)}$ where $p \in M$. note φ is not a chart map here. by chain rule, given $v \in \mathbb{R}^n$, take $\gamma(t) = \varphi^{-1}(p) + tv$. then:

$$\frac{\mathrm{d}}{\mathrm{d}t}(\varphi \circ \gamma)\big|_0 \;=\; D\varphi_p\left(\frac{\mathrm{d}\gamma}{\mathrm{d}t}\bigg|_0\right) \;\implies\; D\varphi\big|_{\varphi^{-1}(p)}(\mathbb{R}^m) \subseteq T_pM$$

to get the reverse inclusion, start with $\gamma : (a,b) \longrightarrow M$ and use $\varphi^{-1} \circ \gamma$. ok idt that was all that important tbh

using directional derivatives, we motivate the defn for an abstract mfld.

Definition 7.1

a **tangent vector** at $p \in M$ is a linear map $v : C^\infty(M) \longrightarrow \mathbb{R}$ of the form:

$$v(f) \;=\; \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} f(\gamma(t))$$

where $\gamma : (a,b) \longrightarrow M$ is some smooth curve with $\gamma(0) = p$.

the idea is to use curves γ to take dir derivatives of f at a pt p. in this sense, each fixed v represents a direction along which we can take the dir derivative of a fn. this identifies it (implicitly, later explicitly) with a particular vector in M which represents the dir more manifestly. this distinct identification separates the direction and its behaviour on fns with the vector in M itself; in this sense, they are really two ways of writing the same thing. tangent spaces drawn in $\mathbb{R}^n$ are thus "geometric representations" of them.

remarks:

1. this is a subset of the inf-dim $\mathrm{Hom}(C^\infty(M), \mathbb{R})$
2. can see it as an equiv rel on curves; that is, v is an equiv class of γ inducing the same linear map. more precisely, curves γ_1, γ_2 are equiv if $\gamma_1(0) = \gamma_2(0) = p$, and $\gamma_1'(0) = \gamma_2'(0)$. v is then the equiv class, and v intuitively represents the vec $\gamma_i'(0)$.

in other words, want to think of them not as vectors but as dir derivatives. we provide an alt coord-wise defn.

Definition 7.2

let $p \in M, (U, \varphi)$ chart arnd p. $v : C^\infty(M) \longrightarrow \mathbb{R}$ a **tangent vector** at p if it is a linear combo of functionals in $\mathrm{Hom}(C^\infty(M), \mathbb{R})$:

$$f \mapsto \sum_{i=1}^{m} a_i \partial_i \left(f \circ \varphi^{-1}\big|_{\varphi(p)} \right)$$

for some $a \in \mathbb{R}^m$, where ∂_i is the i-th partial deriv.

here, $a \in \mathbb{R}^m$ represents the dir vector mapped to coords in $\mathbb{R}^m$. remarks:

1. by defn, is a m-dim'l linear subspace of $\mathrm{Hom}(C^\infty(M), \mathbb{R})$
2. not obv that it doesnt depend on (U, φ)

Theorem 7.3

the two defn's are equiv.

Proof. Source: Primary Source Material

given v, let $\widetilde{\gamma}(t) = \varphi(p) + ta$ a curve $I \to \varphi(U)$ and $\gamma = \varphi^{-1} \circ \gamma$. then:

$$v(f) = \sum_{i=1}^{m} a_i \partial_i \left(f \circ \varphi^{-1}\big|_{\varphi(p)} \right) = \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} (f \circ \varphi^{-1})(\widetilde{\gamma}(t)) = \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} f(\gamma(t))$$

by chain rule, so v of the [first] form. now, if v [of the first form] for some γ w $\gamma(0) = p$, then:

$$v(f) \;=\; \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0}(f \circ \varphi^{-1})(\varphi \circ \gamma(t)) \;=\; \nabla(f \circ \varphi^{-1})\bigg|_{\varphi(p)} \cdot \frac{\mathrm{d}}{\mathrm{d}t}(\varphi \circ \gamma)\bigg|_{t=0}$$

letting $a := \frac{\mathrm{d}}{\mathrm{d}t}(\varphi \circ \gamma)\Big|_{t=0}$, this is then of the second form. ■

ah. so, here we're identifying $v = \sum a_i \partial_i$, or in other words, vectors a with "operators" v. thus, v represents a directional deriv in the direction of a. in the first defn, a is given as the derivative of γ at $t = 0$. <- previous ramblings

ok now onto the txtbk. this is gonna suck.

using the 2nd defn, any chart U arnd p clearly gives an iso $T_pM \simeq \mathbb{R}^m$ by identifying v with a. specifically, if $U \subseteq M$ open, then $T_pU \subseteq C^\infty(U)$ is the subspace spanned by the partials at p:

$$\frac{\partial}{\partial x_1}\bigg|_p, \;\ldots\;, \frac{\partial}{\partial x_m}\bigg|_p$$

this basis identifies $T_pU \simeq \mathbb{R}^m$. coords a_i can then be obtained by taking $v(\pi_i)$.

there is yet another characterization which is coord-free and doesn't rely on curves. first, note that clearly every v satisfies the product rule, also known as the Leibniz rule:

$$v(fg) \;=\; v(f)g(p) + f(p)v(g)$$

evidently, there are many $v \in \mathrm{Hom}(C^\infty(M), \mathbb{R})$ which do not satisfy this property. however, we claim the following:

Theorem 7.4

a linear $v : C^\infty(M) \longrightarrow \mathbb{R}$ defines an elem of T_pM iff it satisfies the product rule.

the pf is rly long and uses "Hadamard's Lemma", i highly doubt its important. im skipping it. anyway, it gives us this third defn:

Definition 7.5

given $v : C^\infty(M) \longrightarrow \mathbb{R}$, $v \in T_pM$ iff:

$$v(fg) \;=\; v(f)g(p) + f(p)v(g)$$

for all f, g.

there is *yet another*, which seemingly involves dual spaces. the book warns to skip it tho, esp if you dont like abstractions. ill read it, but im not taking notes on it.

notice that "velocity vectors" are naturally elements of the tangent space. given a curve γ, then $\gamma'(t_0) \in T_{\gamma(t_0)}M$ is given by:

$$(\gamma'(t_0))(f) \;=\; \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t_0} f(\gamma(t))$$

8 Tangent Maps

Lec 12 - Feb 12 (Week 6)

today: derivatives of smooth maps $f : M \longrightarrow N$. idea: $D_pf : T_pM \longrightarrow T_{f(p)}N$ should generalize the jacobian. recall that for $v \in T_pM$, $v : C^\infty(M) \longrightarrow \mathbb{R}$ is a tangent vec iff it satisfies the product rule, iff:

$$v(g) = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} g(\gamma(t))$$

for a curve γ.

Definition 8.1

we define the tangent map $D_pf : T_pM \longrightarrow T_{f(p)}N$ of f at p as:

$$D_pf(v)(g) \;=\; v(g \circ f)$$

where $g \in C^\infty(N)$, and $g \circ f \in C^\infty(M)$.

this essentially takes a tangent vector $v \in T_pM$ and "pushes" it thru f to get a tangent vector $D_pf(v) \in T_{f(p)}N$. p straightforward.

Proposition 8.2

an equiv defn is that $D_pf([\gamma]) = [f \circ \gamma]$. acc this js shows the above is well-defn'd.

proof:

$$\begin{aligned} D_pf(v)(h) \;=\; v(h \circ f) \;&=\; \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} (h \circ f)(\gamma(t)) \\ &=\; \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} h(f \circ \gamma(t)) \\ &=\; [f \circ \gamma] \end{aligned}$$

notationally, we can write $Df_p(v)$ as f_*v, called the **pushforward** of v. we can denote by f^* a **pullback** by f as in post-composition. if we write $f^*g = g \circ f$ as the pullback of a fn g, $f_*\gamma = f \circ \gamma$ as the pushforward of the curve γ, and $f_*v = (T_pf)(v)$ the pushforward of the tangent vector v, then indeed the defining eq'n becomes $(f_*v)(g) \;=\; v(f^*g)$.

chain rule is written as $D_p(g \circ f)(v) = D_{f(p)}g(D_pf(v))$. proof: symbol push. also, for $f : M \longrightarrow N$, do it on coords. now, we redefine stuff from last wk using tangent spaces.

- the **rank** of f at p is the rank of Df_p
- f is a **submersion** if Df_p is surjective, an **immersion** if it is injective
- $p \in M$ is a **critical pt** of f if $\operatorname{rank} Df_p < \min(m, n)$, a **regular pt** otw
- $q \in N$ is a **regular value** of f if all pts in $f^{-1}(q)$ are regular

given $S \subseteq M$ submfld, the inclusion map ι induces injections $D\iota_p : T_pS \hookrightarrow T_pM$.

example: given $f : M \longrightarrow N$ and reg val q, $S := f^{-1}(q)$ a codim n submfld. we claim that $T_pS = \ker Df_p$. proof: note they have same dim, so enough to prove one inclusion. since $f \circ \iota \equiv q$, take derivatives to get $T_pf(T_p\iota) = 0$.

example: G is a matrix lie grp. lmao okay. this is a subgrp + submfld of $M_n(\mathbb{R})$. then $T_IG \subseteq M_n(\mathbb{R})$ is called the **lie algebra** of G, denoted $\mathfrak{g}$. torture that g on the rack

fact: $\mathfrak{g}$ closed under $[X, Y] = XY - YX$, i.e. the commutator of all X, Y in $\mathfrak{g}$. this is the lie bracket of X, Y.

9 Vector Fields and Tangent Bundles

Lec 12 - Feb 24 (Week 7)

we jump arnd a bit - doing chs 6/9 kinda together.

whats a vector field? morally, a smooth choice of tangent vec at every pt in M. eg, for $M \subseteq \mathbb{R}^n$, tangent vecs live in $T_pM \subseteq \mathbb{R}^n$.

Definition 9.1

a **vector field** is a smooth map $X : M \longrightarrow \mathbb{R}^n$ s.t. $X(p) \in T_pM \subseteq \mathbb{R}^n$.

what does it mean on an abstract mfld? "lets start by cheating". v1: look at charts.

Definition 9.2

a **vector field** X on M is a map $p \mapsto X(p) \in T_pM$, such that for each chart φ, we have $D_p\varphi \circ X : U \longrightarrow \mathbb{R}^n$ is smooth.

this uses the identification $T_pM \simeq \mathbb{R}^n$ induced by chart. but what is that thing between $D_p\varphi \circ X$?

Definition 9.3

the **tangent bundle** TM of M^m is a smooth $2m$-mfld with:

- pt set $\bigsqcup_{p \in M} T_pM$
- (non-max'l) atlas w charts of the form:

$$\left(\bigsqcup_{p \in U} T_pM, (\varphi \circ \pi, D\varphi) \right) \qquad (\varphi \circ \pi, D_p\varphi) : \bigsqcup_{p \in U} T_pM \longrightarrow \mathbb{R}^m \times \mathbb{R}^m$$

for charts (U, φ) in M.

check: π is smooth. remarks:

(1) the tangent bundle comes w a proj $\pi : TM \longrightarrow M$, given by $T_pM \mapsto p$.

(2) for each $p \in M$, the preimg (or fiber) $\pi^{-1}(p)$ is a vspace.

(3) *local triviality:* for each $p \in M$, there's a nbhd $U \ni p$ s.t. the diagram commutes: [TODO: add diagram], where $(D\varphi, \pi)$ is a smooth map and linear on each $\pi^{-1}(q)$.

VI Differential Forms and Integration

12 Covectors and the 1-form

Lec 17 - Mar 12 (Week 9)

... are we reviewing?

to understand VF, need to understand them as:

- *coord-wise*: $X = \sum X_i \partial_i$, where X_i's are smooth fns (in any chart on M)
- *abstract*: a smooth section of the tangent bundle (smooth choice of tangent vec for each point)
- *functional* [or sheaf-theoretic?]: a derivation on $C^\infty(M)$
- *integral*: sth that generates a flow (a 1-param family of diffeos)

we can do sth similar for diff forms.

- *coord-wise*: k-form $\omega = \sum f_I \, \mathrm{d}\omega_{i_1} \wedge \cdots \wedge \mathrm{d}\omega_{i_k}$
- *abstract*: ω is a smooth choice of k-covec at each point (a section of some [the cotangent] vector bundle)
- *functional*: ω a kind of fn $(\mathfrak{X}(M))^k \longrightarrow C^\infty(M)$
- *integral*: a k-form is sth integrable over a k-mfld

Example 12.1

given $f \in C^\infty(M)$, defn its (exterior) differential $\mathrm{d}f : \mathfrak{X}(M) \longrightarrow C^\infty(M)$ by:

$$\mathrm{d}f(X) = X(f)$$

note, $\mathrm{d}f$ is $C^\infty(M)$-linear, i.e.:

$$\mathrm{d}f(gX + hY) = g\mathrm{d}f(X) + h\mathrm{d}f(Y)$$

that is an exercise.

Definition 12.2: Functional

a **1-form** on M is a $C^\infty(M)$-linear map $\mathfrak{X}(M) \longrightarrow C^\infty(M)$. the set ($C^\infty(M)$-alg) of 1-forms is denoted $\Omega^1(M)$.

hey, rmbr dual vspaces? given $\alpha \in V^*$, we can write $\alpha(v)$ as $\langle \alpha, v \rangle$. this emphasizes that if V finite-dim, $(V^*)^* = V$ canonically. that is, we can think of V as a hom $V^* \longrightarrow \mathbb{R}$. note $V^* \simeq V$, but not canonically unless we first fix an inner prod.

Definition 12.3

given basis $e_1, \ldots, e_n$ of V, the **dual basis** for V^* is $e^1, \ldots, e^n$, where $e^i(e_j) = \delta_{ij}$.

cotangent space is defnd precisely as u think. its elems are called **covectors**. in $T_p\mathbb{R}^n$, we have a distinguished basis $\partial_1, \ldots, \partial_n$. its dual basis is written $\mathrm{d}x_1, \ldots, \mathrm{d}x_n$, where:

$$\mathrm{d}x_i(a_1\partial_1 + \cdots + a_n\partial_n) = a_i$$

Definition 12.4

given a lin. map $F : V \longrightarrow W$, theres a **dual map** $F^* : W^* \longrightarrow V^*$ defn'd by $F^*\omega(v) = \omega(Fv)$. [compare this to pullback defn/notation]

given smooth map $f : M \longrightarrow N$, we have a linear map $T_pf : T_pM \longrightarrow T_{f(p)}N$ given by $v \mapsto f_*v = T_pf(v)$. therefore, $T_p^*f : T_{f(p)}^*N \longrightarrow T_p^*M$ is given as $v \mapsto f^*v = T_p^*f(v)$. the "base" map is as a pushforward, i.e. vecs are *covariant*. the dual is as a pullback, i.e. covecs are *contravariant*.

Example 12.5

given $f \in C^\infty(M)$, we can defn a covec $\mathrm{d}_pf \in T_p^*M$ by:

$$\mathrm{d}_pf(v) = v(f) = D_vf$$

in particular, in $\mathbb{R}^n$, we can write $\mathrm{d}_pf = \partial_{x_1}f\mathrm{d}x_1 + \cdots + \partial_{x_n}f\mathrm{d}x_n$. exercise... lol.

Proposition 12.6

given $f \in C^\infty(M, N), g \in C^\infty(N)$, we have $\mathrm{d}_p(f^*g) = (T_p^* f)(\mathrm{d}_p g)$.

Proof.

for any $v \in T_pM$:

$$\begin{aligned}
\langle \mathrm{d}_p(f^*g), v\rangle &= v(f^*g) \\
&= v(g \circ f) \\
&= (T_p f(v))(g) \\
&= \left\langle \mathrm{d}_{f(p)} g, T_p f(v)\right\rangle \\
&= \left\langle (T_p^* f)(\mathrm{d}_{f(p)} g), v\right\rangle
\end{aligned}$$

note $\langle \cdot, \cdot \rangle$ here is evaluation, not an inner prod. ■

Lec 18 - Mar 17 (Week 10)

last time, we talked abt covecs - this is enough for a "vibe" of a defn. what is a 1-form? it is a smooth choice of covec at each pt of a mfld.

Definition 12.7: Coordinate-wise

a map $p \mapsto \alpha_p$ sending pts to covecs is a **diff 1-form** if for every chart (U, φ):

$$\alpha_p = \sum_{i=1}^{m} g_i(\varphi(p))\, \mathrm{d}x_i$$

where $g_i : \varphi(U) \longrightarrow \mathbb{R}$ is smooth.

for instance:

1) $f : M \longrightarrow \mathbb{R}$ as $\mathrm{d}f = \sum \partial_{x_i} f - \mathrm{d}x_i$ on a chart. as an exercise, show this is indep of choice of chart.

2) $y\mathrm{d}x$ on $\mathbb{R}^2$ is not $\mathrm{d}f$ for any f. to see this, note its partials don't commute [why?], or that $y\mathrm{d}x$ depends on y so f depends on y, so $\partial_y f \neq 0$. [what?]

Definition 12.8: Functional

a 1-form is a $C^\infty(M)$-linear $\mathfrak{X}(M) \longrightarrow C^\infty(M)$ given by applying the covec to pts.

Proposition 12.9

let α be a 1-form in the functional sense. then for any VF X, there exists a well-defn'd covec α_p w $p \in M$ s.t. $\alpha(X)(p) = \alpha_p(X_p)$.

Proof.

wts if $X_p = Y_p$, then $\alpha(X)(p) = \alpha(Y)(p)$. therefore, this gives a well-defn'd map $X_p \mapsto \alpha_p(X_p)$. that it's linear, ie a covec, is an exercise. by linearity, enough to show that if $X_p = 0$ then $\alpha(X)(p) = 0$.

indeed, if $X_p = 0$, write $X = \sum f_i Y_i$ where $f_i(p) = 0$ ($Y_i = \partial_i$ on a chart near p, then extend somehow). thus:

$$\alpha(X)(p) \;=\; \sum_i f_i(p)\alpha(Y_i)(p) \;=\; 0$$

■

Proposition 12.10

a) functional 1-forms can be restricted to a form on an open set $U \subseteq M$.

b) the two defns are equiv.

note that this is non-trivial since a VF on U is not always a restriction of a VF on M.

Exercise 12.11

state carefully what the above means, then prove it.

given $g : N \longrightarrow \mathbb{R}$, recall a pullback of g along $f : M \longrightarrow N$ is defined as:

$$f^*g \;=\; g \circ f$$

for covecs, there is an induced map $T_pf : T_pM \longrightarrow T_{f(p)}N$ by $v \mapsto f_*v$, which in turn induces a map $T^*_{f(p)}N \longrightarrow T^*_pM$ by $\nu \mapsto f^*\nu$, where ν is a covec.

now, given a 1-form $\omega \in \Omega^1(N)$, we can define its **pullback** along f by:

$$(f^*\omega)_p \;=\; f^*(\omega_{f(p)})$$

we claim this is indeed a 1-form, i.e. its a smooth choice. indeed, in coords, we have a smooth fn $f : U \longrightarrow V$ where $U, V \subseteq \mathbb{R}^m, \mathbb{R}^n$ resp. if $\omega \in \Omega^1(V)$ given by $\sum g_i \mathrm{d}x_i$, then T_pf has smooth coord fns $\partial_{x_j} f_i$. note:

$$f^*\big|_{f(p)} \;=\; (T_pf)^\intercal$$

so:

$$(f^*\omega)_p \;=\; f^*\Big|_{f(p)}(\omega_{f(p)}) \;=\; f^*\Big|_{f(p)} \sum g_i(f(p))\,\mathrm{d}x_i$$

these are all smooth in p, so we are done.

now, some properties of pullbacks:

- $(f \circ g)^* \;=\; g^* \circ f^*$
- if $g : N \longrightarrow \mathbb{R}$ smooth, $f^*(g\omega) \;=\; f^*g \cdot f^*\omega$
- $f^*(\mathrm{d}g) \;=\; \mathrm{d}(f^*g)$
- in coords:

$$\begin{aligned}(f^*\omega)_p \;=\; f^*\Big|_{f(p)} \sum_i g_i(f(p))\,\mathrm{d}x_i \;&=\; \sum_i g_i(f(p)) f^*\Big|_{f(p)} \mathrm{d}x_i \\ &=\; \sum_i \sum_j g_i(f(p)) \frac{\partial f_i}{\partial x_j}\,\mathrm{d}x_j\end{aligned}$$

Definition 12.12

the **cotangent bundle** of a mfld M is a mfld structure on:

$$T^*M \ := \bigsqcup_{p \in M} T_p^* M$$

with the canonical proj $\pi : T^*M \longrightarrow M$. given (U, φ), we get a chart on T^*M:

$$(\pi^{-1}(U), (\varphi^{-1})^* : \pi^{-1}U \longrightarrow \varphi(U) \times \mathbb{R}^m)$$

here, $\mathbb{R}^m$ is a copy of the cotangent space. note transition maps are smooth, as they are given by:

$$(\varphi \circ \psi^{-1})^* = (T_p(\varphi \circ \psi^{-1}))^\intercal \ : \ \varphi(U \cap V) \times \mathbb{R}^m \longrightarrow \psi(U \cap V) \times \mathbb{R}^m$$

this is just repackaging the coord-wise defn.

NOTE: with tangent bundles, $f : M \longrightarrow N$ defined $Tf : TM \longrightarrow TN$. here, we can only do sth similar when f is a diffeo, as pts go forward but vecs go back. OTOH, if $\omega \in \Omega^1(N)$ is defn'd as $s_\omega : N \longrightarrow T^*N$, then $f^*\omega$ is given by:

$$s_{f^*\omega}(p) \ = \ f^*\big|_{f(p)}(s_\omega \circ f(p))$$

Definition 12.13: Abstract

a 1-form is a section of the cotangent bundle

given $\gamma : I \longrightarrow M$, $\alpha \in \Omega^1(M)$, take $\gamma^*\alpha \in \Omega^1(I)$ as $\gamma^*(\alpha) = f(t)\ \mathrm{d}t$ for some $f(t)$.

with this, we can define the integral of a 1-form.

Definition 12.14

we define:

$$\int_\gamma \alpha \;=\; \int_I \gamma^*\alpha \;=\; \int_a^b f(t)\,\mathrm{d}t$$

where $I = [a, b]$. we'll take smoothness on closed intervals in the extension way.

Proposition 12.15

if $\alpha = \mathrm{d}f$ for some $f : M \longrightarrow \mathbb{R}$, then:

$$\int_\gamma \alpha \;=\; f(\gamma(b)) - f(\gamma(a))$$

Proof.

we have that $\gamma^*\mathrm{d}f \;=\; \mathrm{d}(\gamma^* f) \;=\; \mathrm{d}(f \circ \gamma) \;=\; \partial_t(f \circ \gamma)\mathrm{d}t$. apply usual ftc. ■

for instance, $y\mathrm{d}x$ is not $\mathrm{d}f$, i.e. an *exact form*, since by integrating counterclockwise on $\partial[0,1]^2$, we get:

$$\int_\gamma y\,\mathrm{d}x \;=\; 0 + 0 + \int_0^1 1(-\mathrm{d}t) + 0 \;=\; 1 \;\neq\; 0 \;=\; f(0,0) - f(0,0)$$

Proposition 12.16

integrating is invariant wrt reparametrization:

$$\int_\gamma \alpha \;=\; \pm \int_{\gamma\circ\kappa} \alpha$$

where $\kappa : [c, d] \longrightarrow [a, b]$ is a diffeo, and $\pm$ depends on orientation.

Proof.

we have:

$$\int_{\gamma\circ\kappa} \alpha \;=\; \int_c^d (\gamma\circ\kappa)^*\alpha \;=\; \int_c^d \kappa^*\gamma^*\alpha \;=\; \pm\int_{\kappa(c)}^{\kappa(d)} \gamma^*\alpha$$

after a substitution. ■

recall α is *exact* if its $\mathrm{d}f$ for some $f \in C^\infty$. in this case, $\int_\gamma \alpha$ depends only on endpts.

fact: this is an iff. if $\int_\gamma \alpha$ depends only on endpts, then defn $f(p) = \int_\gamma \alpha$ where γ is any curve p_0 to p (where we can choose p_0). to ensure f smooth and $\mathrm{d}f = \alpha$, do a coord calculation. ofc, this is iff $\int_\gamma \alpha = 0$ for γ a loop.

Example 12.17

let $M = S^1, \alpha = \mathrm{d}\theta$. then:

$$\int_{S^1} \mathrm{d}\theta \;=\; \int_\gamma \mathrm{d}\theta \;=\; 2\pi$$

where γ inj on (a, b) and $\gamma(a) = \gamma(b)$. thus, $\mathrm{d}\theta$ is not an exact form.

13 General k-forms

Lec 19 - Mar 19 (Week 10)

last time, we talked abt integrating 1-forms:

$$\int_\gamma \alpha \;=\; \int_a^b \gamma^*\alpha \;=\; \int_a^b \alpha_{\gamma(t)}(\gamma_*(\partial_t))\,\mathrm{d}t$$

this is indep of parametrization of γ, since $\alpha_{\gamma(t)}(rv) = r\alpha_{\gamma(t)}(v)$. integral remains the same by substitution. that $\alpha(X+Y) = \alpha(X) + \alpha(Y)$ doesnt matter for this computation.

we want k-forms to be sth we integrate over k-dim submflds. (in)formally, this is a map:

$$\omega : \underbrace{T_pM \times \cdots \times T_pM}_{k \text{ times}} \longrightarrow \mathbb{R}$$

then, for $F : U \subseteq \mathbb{R}^k \longrightarrow M$, we can define:

$$\int_{F(U)} \omega \;=\; \int_U \omega_{F(x)}(F_*\partial_1, \ldots, F_*\partial_k)\, \mathrm{d}x_1 \cdots \mathrm{d}x_k$$

we *want* invariance under reparametrization $G : U \longrightarrow V$ diffeo, that is:

$$= \int_{G(U)} \omega_{F(G^{-1}(y))} \left((F \circ G^{-1})_*\partial_1, \ldots, (F \circ G^{-1})_*\partial_k\right)\, \mathrm{d}y_1 \cdots \mathrm{d}y_k$$

which, by change of variables from 257, is given as:

$$= \pm \int_U \omega_{F(x)} \left((F \circ G^{-1})_*\partial_1, \ldots, (F \circ G^{-1})_*\partial_k\right) \left|\det TG^{-1}\right| \mathrm{d}x_1 \cdots \mathrm{d}x_k$$

so what do we acc need? recall the following fact: if V is k-dim'l, then given any map $\omega : V \times \cdots \times V \longrightarrow \mathbb{R}$ which is multilinear and skew-symmetric, then we have that $\omega(v_1, \ldots, v_k) = C \cdot \det[v_1 \;\cdots\; v_k]$. thus, if $\omega : V^k \longrightarrow \mathbb{R}$ for n-dim'l V is multilinear and alternating, its restriction to *any* k-dim subspace behaves like a det.

denoting these as alternating (or skew-sym) k-tensors, our vague defn is that a k-form $\omega \in \Omega^k(M)$ is a smooth choice of alt k-tensors $\omega_p : T_pM \times \cdots \times T_pM \longrightarrow \mathbb{R}$.

Definition 13.1: Functional

a **k-form** is an alternating multilinear map: $\omega : \underbrace{\mathfrak{X}(M) \times \cdots \times \mathfrak{X}(M)}_{k \text{ times}} \longrightarrow C^\infty(M)$.

for instance, $\omega \in \Omega^n(\mathbb{R}^n)$ always looks like:

$$\omega(X_1, \ldots, X_n)(p) \;=\; f(p) \det[(X_1)_p \;\cdots\; (X_n)_p]$$

for some $f \in C^\infty(\mathbb{R}^n)$. what do alt k-tensors look like in general?

Definition 13.2

let $I = \{i_1 < \cdots < i_k\} \subseteq \{1, \ldots, n\}$. define $\mathrm{d}x_I = \mathrm{d}x_{i_1} \wedge \cdots \wedge \mathrm{d}x_{i_k}$ to be the alt k-tensor which satisfies:

$$\mathrm{d}x_I(\partial_{i_1}, \ldots, \partial_{i_k}) = 1 \qquad \mathrm{d}x_I(p_j, \text{ anything else }) = 0$$

where $j \notin I$. note this uniquely defns an alt k-tensor.

here, the $\wedge$ is simply a symbol (for now). intepretation: take $v_1, \ldots, v_k$ and compute $\det[\pi_I v_1 \ \cdots \ \pi_I v_k]$, where π_I projects to $\langle \partial_{i_1}, \ldots, \partial_{i_k} \rangle$.

Lemma 13.3: Coordinate-wise

every alt k-tensor $\omega : (\mathbb{R}^n)^k \longrightarrow \mathbb{R}$ can be expressed uniquely as:

$$\omega = \sum_{\substack{I \subseteq \{1,\ldots,n\} \\ |I|=k}} \omega_I \, \mathrm{d}x_I \qquad \omega_I \in \mathbb{R}$$

Proof.

note $\omega_I = \omega(\partial_{i_1}, \ldots, \partial_{i_k})$, check that $\omega(v_1, \ldots, v_k)$ determined by these. (what?) ■

Proposition 13.4

any k-form on $U \subseteq \mathbb{R}^n$ is given by:

$$\omega = \sum_{\substack{I \subseteq \{1,\ldots,n\} \\ |I|=k}} f_I(p) \, \mathrm{d}x_I$$

for $f_I : U \longrightarrow \mathbb{R}$ smooth. any k-form on M can be written this way on charts. in particular, $\omega(X_1, \ldots, X_k)(p) = \omega_p((X_1)_p, \ldots, (X_k)_p)$ for some well-defined alt k-tensor ω_p.

pf is same as for VF and 1-forms.

Lec 20 - Mar 24 (Week 11)

tdy, we look at operations on forms:

- wedge product
- exterior derivative
- pullback along fns
- integration over submflds (time permitting)

we start w the wedge product. this is an operation $\Omega^k(M) \times \Omega^\ell(M) \longrightarrow \Omega^{k+\ell}(M)$. as warmup, lets look at the case where $k = \ell = 1$.

let $\alpha, \beta \in \Omega^1(M)$. does $\alpha \wedge \beta(v_1, v_2) = \alpha(v_1)\beta(v_2)$ work? not necessarily, since it may not be alternating. we can instead force this by instead setting:

$$\alpha \wedge \beta(v_1, v_2) \ := \ \alpha(v_1)\beta(v_2) - \beta(v_1) - \alpha(v_2)$$

for instance, on $\mathbb{R}^2$ with $\alpha = \mathrm{d}x, \beta = \mathrm{d}y$, sps $v_i = (x_i, y_i)$. then:

$$\alpha \wedge \beta(v_1, v_2) \ = \ x_1 y_2 - x_2 y_1 \ = \ \det \begin{bmatrix} x_1 & x_2 \\ y_1 & y_2 \end{bmatrix} \ = \ \mathrm{d}x \wedge \mathrm{d}y(v_1, v_2)$$

more generally, on $\mathbb{R}^n$ with $\alpha = \mathrm{d}x_1, \beta = \mathrm{d}x_2$, we have that $\alpha \wedge \beta(\partial_i, \partial_j) = 0$ unless $i, j = 1, 2$ in some order, $\alpha \wedge \beta(\partial_1, \partial_2) = 1$, and $\alpha \wedge \beta(\partial_2, \partial_1) = -1$.

Definition 13.5

sps $\alpha \in \Omega^k(M), \beta \in \Omega^\ell(M)$. then the **wedge product** $\alpha \wedge \beta \in \Omega^{k+\ell}(M)$ is:

$$\begin{aligned} \alpha \wedge \beta(v_1, \ldots, v_{k+\ell}) \ &= \ \frac{1}{k!\ell!} \sum_{\sigma \in S_{k+\ell}} \operatorname{sign}(\sigma) \alpha(v_{\sigma(1)}, \ldots, v_{\sigma(k)}) \beta(v_{\sigma(k+1), \ldots, v_{\sigma(k+\ell)}}) \\ &= \ \sum_{\sigma \in \Sigma_{k,\ell}} \operatorname{sign}(\sigma) \alpha(v_{\sigma(1)}, \ldots, v_{\sigma(k)}) \beta(v_{\sigma(k+1)}, \ldots, v_{\sigma(k+\ell)}) \end{aligned}$$

where $\Sigma_{k,\ell}$ represents the "(k, ℓ) shuffles", that is, permutations which preserve the independent ordering of $[k]$ and $[k + \ell] \setminus [k]$.

in particular, $\mathrm{d}x_I = \mathrm{d}x_{i_1} \wedge \cdots \wedge \mathrm{d}x_{i_k}$ for $i_1 < \cdots < i_k$. that is, our earlier "symbol" was in fact the right thing.

some properties of the wedge product:

i) *bilinear*: [check that] $f(\alpha_1 + \alpha_2) \wedge \beta \ = \ f\alpha_1 \wedge \beta + \alpha_2 \wedge \beta$

ii) *graded commutative*: $\beta \wedge \alpha \ = \ (-1)^{k\ell}\alpha \wedge \beta$ (here, $k\ell$ is the sign of the permutation $1, \ldots, k, k+1, \ldots, k+\ell \to k+1, \ldots, k+\ell, 1, \ldots, k$)

iii) *associative*: yk this. it just works

Corollary 13.6

every form is locally a sum of forms of form $f_0 \mathrm{d}f_1 \wedge \cdots \wedge \mathrm{d}f_k$, where $f_0, \ldots, f_k$ are smooth fns

Proof.

$f_0 \mathrm{d}f_1 \wedge \cdots \wedge \mathrm{d}f_k$ in local coords and $\mathrm{d}x_i = \mathrm{d}\pi_i$ ■

objection: coord fns dont extend to M! whats the soln? pick $K \subseteq U$ cpt, $V \ni p$ open in K, $\varphi : M \longrightarrow \mathbb{R}$ w supp K, 1 on V. then we use $f_i = \varphi x_i$ which extend to M, and:

$$\omega \ = \ \sum_{\substack{|I|=k \\ I \subseteq \{1,\ldots,n\}}} \omega_I \mathrm{d}f_{i_1} \wedge \cdots \wedge \mathrm{d}f_{i_k}$$

moving on to the exterior derivative: $\Omega^k \longrightarrow \Omega^{k+1}$. this is a generalization of $f \mapsto \mathrm{d}f$.

a defn/thm:

Theorem 13.7

theres a unique coll'n of lin maps $\Omega^k(M) \longrightarrow \Omega^{k+1}(M)$ for each k s.t.:

i) for $f \in \Omega^0(M) = C^\infty(M)$, $\mathrm{d}f \in \Omega^1$ is the differential

ii) $\mathrm{d}(\mathrm{d}f) = 0$ for all $f \in C^\infty(M)$

iii) the **graded product rule** holds:

$$\mathrm{d}(\alpha \wedge \beta) \;=\; \mathrm{d}\alpha \wedge \beta + (-1)^{\deg\alpha}\alpha \wedge \mathrm{d}\beta$$

iv) (bonus!) $\mathrm{d}(\mathrm{d}\alpha) = 0$ for all $\alpha \in \Omega^k$

general idea/outline of pf: assume existence, prove uniqueness, then derive a formula.

1. d is *local*: $(\mathrm{d}\alpha)\big|_U$ only depends on $\alpha\big|_U$ if and only if $\alpha\big|_U = 0 \implies (\mathrm{d}\alpha)\big|_U = 0$. uhhh idk what hes doing ngl. some sort of bump fn shenanigans idk what f is i missed it

   by the properties:
   $$0 = \mathrm{d}(f\alpha) = \mathrm{d}f \wedge \alpha + f\mathrm{d}\alpha$$
   since $\mathrm{d}f = 0$ and $f = 1$ at p, so $(\mathrm{d}\alpha)\big|_p = 0$. we can do this for all $p \in U$. thus, its enough to see what happens on charts.

2. by properties:
   $$\mathrm{d}(f_0\mathrm{d}f_1 \wedge \cdots \wedge \mathrm{d}f_k) \;=\; \mathrm{d}f_0 \wedge \mathrm{d}f_1 \wedge \cdots \wedge \mathrm{d}f_k$$
   this gives us the formula in coords:
   $$\mathrm{d}(f\mathrm{d}x_I) \;=\; \mathrm{d}f \wedge \mathrm{d}x_I$$
   this gives the uniqueness and forms property.

3. now, show this formula actually works, i.e., satisfies the properties.

4. finally, show existence, ie the formula is consistent over diff charts. to see this, use uniqueness to show theyre the same on overlaps.

Definition 13.8

a form $\omega \in \Omega^k(M)$ is **closed** if $\mathrm{d}\omega = 0$, and **exact** if $\omega = \mathrm{d}\alpha$.

for instance, all m-forms on M^m are closed. the form we showed last time, $\mathrm{d}\theta \in \Omega^1(S^1)$, is closed but not exact. OTOH, every exact form is closed.

Definition 13.9

the **de Rham cohomology** of M is the vector space given by:

$$H^k(M) \;=\; \frac{\text{closed forms}}{\text{exact forms}}$$

facts: (hard!)

i) $H^k(M)$ is finite-dim if M is cpt

ii) $H^k(M)$ only depends on the topology of M, and not its diffeo type

ok moving on to pullbacks of forms.

Definition 13.10

given $F : M \longrightarrow N$, $\alpha \in \Omega^k(N)$, we define:

$$F^*\alpha(v_1, \ldots, v_k) \;=\; \alpha(F_*v_1, \ldots, F_*v_k)$$

properties:

i) is in fact a diff form (varies smoothly)

ii) preserves wedge: $F^*(\alpha \wedge \beta) \;=\; F^*\alpha \wedge F^*\beta$

iii) preserves d: $F^*(\mathrm{d}\alpha) \;=\; \mathrm{d}F^*\alpha$

Proof.

1. pointwise, (ii) holds:

$$\begin{aligned} F^*(\alpha \wedge \beta)(v_1, \ldots, v_k) &= (\alpha \wedge \beta)(F_*v_1, \ldots, F_*v_k) \\ &= \sum_{\sigma \text{ shuffles}} \alpha(F_*v_{\sigma(1)}, \ldots, F_*v_{\sigma(k)})\beta(\cdots) \\ &= F^*\alpha \wedge F^*\beta(v_1, \ldots, v_k) \end{aligned}$$

2. $\mathrm{d}, \wedge, F^*$ are all $\mathbb{R}$-lin maps + local, so we only have to show this for $f_0 \mathrm{d}f_1 \wedge \cdots \wedge \mathrm{d}f_k$

3. by step 1:

$$F^*(f_0 \mathrm{d}f_1 \wedge \cdots \wedge \mathrm{d}f_k) \;=\; F^* f_0 F^* \mathrm{d}f_1 \wedge \cdots \wedge F^* \mathrm{d}f_k$$

we also know pullbacks of 1-forms are 1-forms, so we get (i). we also also know $F^*(\mathrm{d}f) = \mathrm{d}F^* f$, so we get (iii).

■

in particular, we see from last time that for $F : \mathbb{R}^k \longrightarrow \mathbb{R}^k$, every k-form ω is:

$$\omega \;=\; f \mathrm{d}x_1 \wedge \cdots \wedge \mathrm{d}x_k$$

for some f. thus:

$$\begin{aligned} F^*\omega(\partial_1, \ldots, \partial_k) &= \omega(F_*\partial_1, \ldots, F_*\partial_k) \\ &= \omega(\partial_1 F, \ldots, \partial_k F) \\ &= \det(TF)\omega(\partial_1, \ldots, \partial_k) \end{aligned}$$

ie, $F^*\omega\big|_p \;=\; \det(T_pF)\omega\big|_{F(p)}$.

14 Integration of forms

moving on to integration of diff forms

for k-forms on $\mathbb{R}^k$, if f has cpt sppt in U, we can defn:

$$\int_U \omega \;=\; \int_{\mathbb{R}^k} f(x_1, \ldots, x_k)\, \mathrm{d}x_1 \cdots \mathrm{d}x_k$$

from last time, we see that $\int_{F(U)} \omega = \pm \int_U F^*\omega$ by cov, where $F : U \longrightarrow F(U)$ diffeo.

for k-forms on a k-mfld: sps $\omega \in \Omega^k(M^k)$ has cpt sppt. then we can find charts $U_1, \ldots, U_r$ st ω supp on $\bigcup U_i$, and can write:

$$\omega = \rho_1\omega + \cdots + \rho_r\omega$$

where $\rho_i : M \longrightarrow \mathbb{R}$ is a parition of unity w sppt U_i. then by defn:

$$\int_M \omega \;=\; \sum_{i=1}^{r} \int_{\varphi_i(U_i)} (\varphi_i^{-1})^*(\rho_i \omega)$$

where we choose φ_i's to have compat orientations, ie $\det\big(T(\varphi_i \circ \varphi_j^{-1})\big) > 0$ for all i, j.

Lec 21 - Mar 26 (Week 11)

last time, we talked abt integrating k-forms on k-mflds. an important note is that the defn from last time is indep of coords. we can do sth similarly for all forms.

Definition 14.1

given $\omega \in \Omega^k(M^m)$, k-dim'l oriented submfld $\iota : S \hookrightarrow M$, we can define:

$$\int_S \omega \;=\; \int_S \iota^* \omega$$

note on the left, $S \subseteq M$ and $\omega \in \Omega^k(M)$, but on the right, S is a mfld in and of itself and $\iota^*\omega \in \Omega^k(S)$.

goal for tdy: prove generalized stokes (we have 30 mins left in class). here's the statement.

Theorem 14.2: Generalized Stokes' Theorem

let $A \subseteq M^m$ be a *bounded region* with *boundary* ∂A. then for any $\alpha \in \Omega^{m-1}(M)$ with $\operatorname{supp}(\alpha) \cap A$ cpt, we have:

$$\int_{\partial A} \omega \;=\; \int_A \mathrm{d}\omega$$

remark: M may be a submfld of sth higher-dim! other remark: we literally havent talked abt bdries lmao

Example 14.3

i) sps $\alpha \in \Omega^2(\mathbb{R}^3)$, A some bounded region in $\mathbb{R}^3$. then this yields the divergence theorem.

ii) sps $\alpha \in \Omega^1(\mathbb{R}^3)$, A some bounded surface in $\mathbb{R}^3$. then this yields the classical stokes thm.

iii) sps $\alpha \in \Omega^0(M) = C^\infty(M)$. then this yields the fundamental thm of calculus.

gave 2 exs w pictures. oh well

Corollary 14.4

(more of a special case) if α closed, then:

$$\int_{\partial A} \alpha \;=\; 0$$

Definition 14.5

sps M oriented. we say $D \subseteq M$ is a **region with smooth boundary** if $\exists f : C^\infty(M)$ with 0 as a reg val and $D = f^{-1}(t \leq 0)$. we write $\partial D = f^{-1}(0)$ and $\text{int}(D) = f^{-1}(t < 0)$.

fix a special atlas covering D:

- cvr ∂D by submfld charts (U, φ) compat w orientation of M, ie:

$$\varphi(U \cap \partial D) \;=\; \varphi(U) \cap \{x_1 = 0\} \qquad \varphi(U \cap \text{int}D) \;=\; \varphi(U) \cap \{x_1 < 0\}$$

  after possibly flipping sign of x_1

- cvr interior by more oriented charts

this gives an oriented atlas on D.

Proposition 14.6

when restricting to $\{x_1 = 0\}$, we get an oriented atlas on ∂D

Proof.

let $\tau : \varphi_i(U_i \cap U_j) \longrightarrow \varphi_j(U_i \cap U_j)$ be the transition map. given $x = (0, x_2, \ldots, x_m)$, consider $T_x\tau$: the top left is > 0 since "away from D" is preserved, the rest of the first row is 0 [cz bdry is preserved? idk], and ignoring the first column, the remaining matrix is made up of the jacobian of the transition map btwn charts on ∂D.

we know $\det T_x\tau > 0$, so det(that other thing) is +ve, so the atlas $\left(U\big|_{\partial D}, \varphi\big|_{\partial D}\right)$ is oriented. ■

note: the orientation we put on ∂D is engineered to avoid a minus sign.